

World Happiness Database Project

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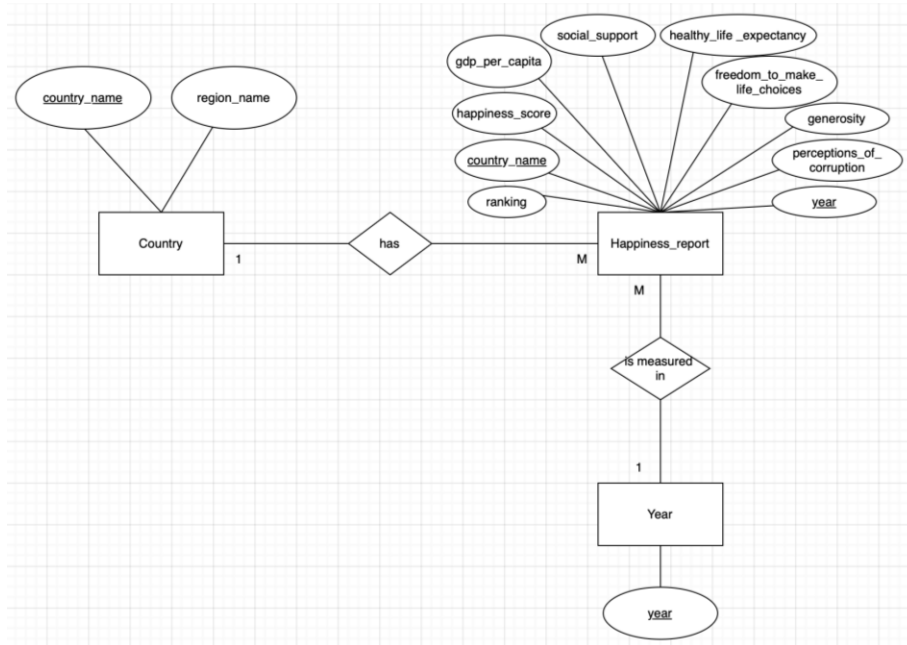
Lejla Kovac (Reproducibility aspects)

Research Problem and Motivation

Which factors have the greatest influence on the global happiness score between 2015 and 2024?

- The factors we look at in the database: GDP (BIP), social support, healthy life expectancy, freedom to make life choices, generosity, perception of corruption.
- [World Happiness 2015-2024](#) Database
- Why store it in relational database: These variables are structured, logically connected rather than independent (M:country-1:Happiness Score).

Data Modeling and data Ingestion



Entities:

- Country: country_name(PK), region_name
- Year: year(PK)
- Happiness_report: country_name(PK,FK), year(PK,FK), ranking, happiness_score, gdp_per_capita, social_support, healthy_life_expectancy, freedom_to_make_life_choices, generosity, perceptions_of_corruption

Data Ingestion Process:

- create 3 tables: country, year, happiness_report
- Imported the CSV file into temporary table to store all columns as TEXT
- Data cleaned and converted into correct types
- Country and Year populated using SELECT DISTINCT and for region MIN(region_name)
- Converted TEXT into NUMERIC, because decimal commas appear in the source data
- Happiness_report populated last to satisfy foreign key constraints
- Normalized schema again

Database queries and data handling

- The first query computes the Pearson correlation of happiness factors. The second one computes the same thing but focusing on Austria (We picked one country 'randomly' to see if the correlations are similar).
- Then the third query shows the average happiness and social support per year globally, along with their correlation, to visualize trends over time.
- Query no. 4 retrieves yearly happiness and social support for Austria, allowing comparison of trends within the country. (we picked the same country again to showcase the correlation, but with only 1 data per year, it is not statistically correct to compute it so it's just to show the trend lines)

Presentation of the results

Query 1 & 2: Pearson correlations between happiness score and other variables

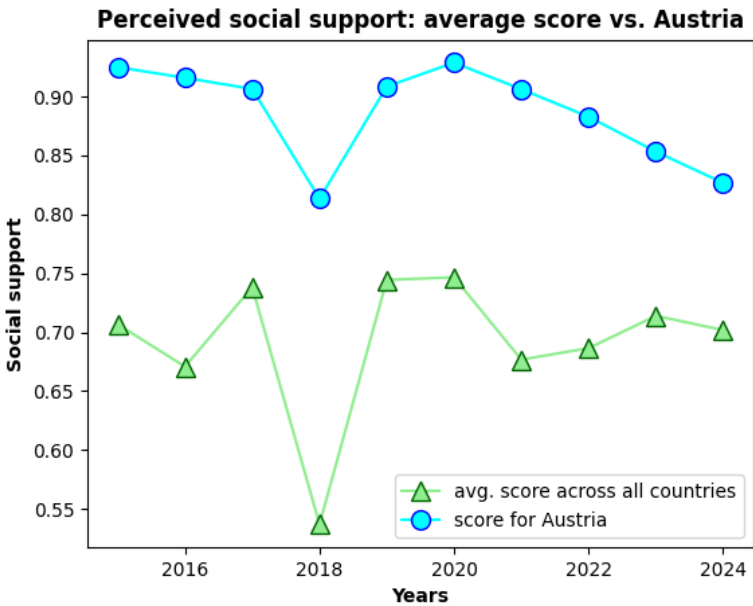
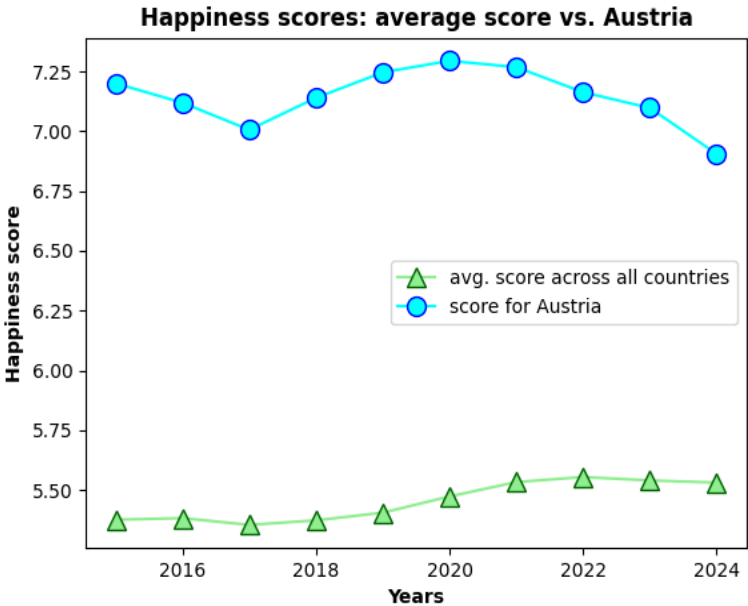
Factor	Overall	Austria
GDP per capita	0.631*	-0.142
Social Support	0.744*	0.575
Healthy Life Expectancy	0.66*	-0.035
Freedom to make life choices	0.591*	0.489
Generosity	0.106*	-0.423
Corruption	0.071*	0.034

Highest correlation globally and Austria: Social Support with Happiness Score (a higher level of perceived social support also leads to a higher happiness score).
*p_value<0.05



Scatter Plot with the Social Support values of all countries (our variable with the highest correlation of .744) to show the correlation of these two variables.

Query 3 & 4: Comparison of average vs. Austrian scores



Interpretation:

The perceived social support has the highest impact on the happiness score, globally as well as in Austria. In comparison to other countries, Austria has a high level of happiness and also a high value of perceived social support over the years.

Reproducibility aspects

- **Open and stable data source:** World Happiness Report
- **Direct dataset reference:** [World Happiness 2015-2024](#)
- **Explicit schema assumptions:** country, year, and indicator variables
- **Deterministic preprocessing:** no stochastic steps or manual intervention
- **Reusable SQL appendices:** Query_1.sql - Query_4.sql
- **Sequential execution order:** queries must be run in defined order
- **Tool independence:** compatible with standard SQL databases
- **Result traceability:** each output can be linked to a specific query
- **Reproduction workflow:** data import → query execution → result generation
- **Reproducibility constraints:** dataset versioning and execution environment

GitHub: <https://github.com/24katharinaLienhart/Databases-Group-project>